

Prefetch-Aware Cache Management: A Verilog-Based RTL Implementation

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Abstract—This paper presents the RTL design and simulation of a prefetch-aware cache management system targeting improved Average Memory Access Time (AMAT) in processor memory hierarchies. The system integrates a 4-way set-associative L1 cache with a hardware stride prefetcher, connected through a priority arbiter. Implemented entirely in Verilog and verified using Vivado XSim, the design demonstrates a 40% reduction in AMAT—from 4.0 cycles (baseline, no prefetcher) to 2.4 cycles (with stride prefetcher enabled)—across a 20-access sequential workload. Four testbenches covering cold miss behavior, LRU replacement, and performance comparison validate the correctness and effectiveness of each module.

Index Terms—Cache memory, stride prefetcher, set-associative cache, LRU replacement, AMAT, RTL design, Verilog.

I. INTRODUCTION

The memory wall—the growing disparity between processor speed and DRAM latency—remains a primary performance bottleneck in modern computer architectures. L1 caches are the first line of defense, but their effectiveness depends critically on the access patterns presented by executing programs. For workloads with regular, strided access patterns (e.g., array traversals, matrix operations), hardware prefetching can preload cache lines before the CPU demands them, converting costly cache misses into hits.

This work designs, implements, and verifies a complete prefetch-aware cache management system at the register-transfer level (RTL) in Verilog. The system comprises four key modules:

- **RAM** – 128-block main memory (16 words/block, 32 bits/word).
- **cache_4way** – 4-way set-associative L1 cache with LRU replacement.
- **stride_prefetcher** – 4-entry hardware stride detection engine.
- **arbiter** – Combinational priority multiplexer (CPU > prefetcher).
- **top** – Integration module with `prefetch_enable` control input.

The primary objective is to demonstrate a measurable AMAT improvement attributable solely to the prefetcher, using matched testbenches (with and without prefetching) as a controlled experiment.

II. SYSTEM ARCHITECTURE

A. Address Organisation

The 11-bit address is partitioned as follows:

$$\underbrace{\text{addr}[10 : 8]}_{\text{tag (3 bits)}} \quad \underbrace{\text{addr}[7 : 4]}_{\text{set index (4 bits)}} \quad \underbrace{\text{addr}[3 : 0]}_{\text{block offset (4 bits)}}$$

This gives 16 sets, 4 ways per set, and 16 words per cache block. The RAM is addressed identically, organised as 128 lines $\times$ 16 words $\times$ 32 bits.

B. Cache Module (*cache_4way*)

The cache implements a 4-state Mealy FSM:

S0 (IDLE) $\rightarrow$ **S1 (CHECK)** $\rightarrow$ **S2 (FETCH)** $\rightarrow$ **S3 (FILL)**
 $\rightarrow$ **S0**

- **S0:** Waits for `cs` && `rd`. The `busy` output is de-asserted.
- **S1:** Tag comparison across all 4 ways. On a hit, data is returned and LRU counters updated in the same cycle. On a miss, transitions to S2.
- **S2:** Issues RAM read request. Victim way selection uses an invalid-slot priority: a free (invalid) way is chosen first; only if all ways are valid is LRU eviction applied.
- **S3:** Writes the 512-bit RAM block into the selected cache way and presents the requested word at `data_out`.

LRU Policy. Each way holds a 2-bit age counter (0 = MRU, 3 = LRU). On a hit, only ways *more recently used* than the hit way are aged; the hit way is set to 0. On a fill, similarly, only valid peers are aged.

Address latching. A critical design point: the arbiter de-asserts `cache_cs/cache_rd` as soon as the prefetcher's `prefetch_req` is cleared. To prevent the FSM from operating on a stale live address, the address bits (`addr_tag`, `set_index`, `block_off`) are derived from the combinational `addr` input which, in S2/S3, is held stable by the arbiter for the duration of the transaction.

C. Stride Prefetcher (*stride_prefetcher*)

The prefetcher maintains a 4-entry table. Each entry tracks:

- `prev_addr`: last address accessed in this stream.
- `stride`: signed 12-bit detected stride.
- `state`: one of INITIAL, TRANSIENT, STEADY.
- `valid`: entry occupancy flag.

State transitions follow the standard 3-phase stride detection protocol:

- 1) **INITIAL** $\rightarrow$ **TRANSIENT**: First stride δ observed.
- 2) **TRANSIENT** $\rightarrow$ **STEADY**: Stride confirmed (δ matches stored value).
- 3) **STEADY**: Issues `prefetch_req` with `prefetch_addr = cpu_addr + stride` for each confirmed stride access.

Edge detection. A `prev_cs` register ensures the prefetcher only samples on the *rising edge* of `cs` && `rd` (condition: `cs && rd && !prev_cs`). Without this, the same address is sampled repeatedly during the 4-cycle cache miss, causing a spurious stride-of-zero to be learned.

Two-phase handshake. A `wait_for_busy` flag prevents `prefetch_req` from being cleared prematurely. The sequence is: (1) assert `prefetch_req` and `wait_for_busy`; (2) wait until cache asserts `busy` (it has accepted the prefetch); (3) wait until `busy` deasserts; (4) clear `prefetch_req`.

D. Arbiter (*arbiter*)

A purely combinational priority mux with the rule: CPU > prefetcher > idle. The CPU always preempts any pending prefetch request when it asserts `cpu_cs` && `cpu_rd`.

E. Top-Level Integration (*top*)

A `prefetch_enable` input selects between two signal paths:

- `prefetch_enable = 1`: arbiter drives the cache (full system).
- `prefetch_enable = 0`: CPU drives the cache directly (baseline).

This allows identical testbench code to compare both configurations without any structural change.

III. VERIFICATION METHODOLOGY

Four testbenches were developed. All use a common `send_request` task that (a) asserts `cs/rd`, (b) waits for `busy` to go high (FSM entered S1), (c) samples the hit signal, (d) waits for `busy` to fall, and (e) immediately deasserts `cs/rd` to prevent FSM re-triggering.

A. Testbench 1: Cold Miss and Repeated Hit Test

Purpose. Verify basic hit/miss detection and LRU update without the prefetcher.

Access pattern. 30 accesses: 20 unique addresses (cold misses) followed by 10 repeat accesses (warm hits).

Expected. 20 misses + 10 hits.

Observed (Fig. 1a). `total_accesses = 0x1E = 30`, `total_hits = 0x0A = 10`, `total_misses = 0x14 = 20`. ✓

B. Testbench 2: LRU Eviction Test

Purpose. Verify correct LRU victim selection when all 4 ways in a set are valid.

Access pattern. Fill a single set with 4 different tags (ways 0–3). Access way 0, way 1, way 2 (updating LRU order). Install a 5th tag—this must evict way 3 (LRU). Re-verify the surviving ways are still readable.

Observed (Fig. 1b). `total_accesses = 0x20 = 32`, `total_hits = 8`, `total_misses = 0x18 = 24`. Eviction targeted the correct LRU way. ✓

C. Testbench 3: Baseline Performance (No Prefetcher)

Purpose. Establish the baseline AMAT for 20 sequential accesses with `prefetch_enable = 0`.

Access pattern. Addresses 0x003, 0x013, 0x023, ..., 0x133 (stride = 0x010, crossing 20 distinct cache sets and tags).

Expected. All 20 accesses are cold misses; $AMAT = 2 + (1.0 \times 2) = 4.0$ cycles.

Observed (Fig. 1c). `total_accesses = 20`, `total_hits = 0`, `total_misses = 20`, $AMAT = 4.0$ cycles. ✓

D. Testbench 4: Prefetcher-Enabled Performance

Purpose. Measure AMAT improvement with `prefetch_enable = 1` under the same 20-access workload.

Prefetcher learning sequence.

- Access 1 (0x003): Entry allocated, state = INITIAL. Miss.
- Access 2 (0x013): Stride 0x010 observed, state = TRANSIENT. Miss.
- Access 3 (0x023): Stride confirmed, state = STEADY. Miss.
- Access 4 (0x033): STEADY – prefetch 0x043 issued. Miss.
- Accesses 5–20: Each CPU access finds the next address already in cache. Hit.

Expected. 4 misses, 16 hits; $AMAT = 2 + (0.2 \times 2) = 2.4$ cycles.

Observed (Fig. 1d). `total_accesses = 20`, `total_hits = 16 (0x10)`, `total_misses = 4`, `total_prefetch_reqs = 17 (0x11)`, $AMAT = 2.4$ cycles. ✓

IV. RESULTS AND ANALYSIS

A. Performance Summary

Table I summarises the comparative performance measured in Testbenches 3 and 4.

B. AMAT Calculation

The AMAT model used is:

$$AMAT = T_{hit} + \text{Miss Rate} \times T_{miss \text{ penalty}}$$

where $T_{hit} = 2$ cycles (S0 $\rightarrow$ S1 in one clock, data ready next) and $T_{miss \text{ penalty}} = 2$ cycles (S2 + S3).

TABLE I: Performance Comparison: Without vs. With Stride Prefetcher

Metric	No Prefetcher	With Prefetcher
Total Accesses	20	20
Cache Hits	0	16
Cache Misses	20	4
Prefetch Requests	–	17
Hit Rate	0%	80%
Miss Rate	100%	20%
AMAT (cycles)	4.0	2.4
AMAT Reduction	–	40%

Without prefetcher:

$$\text{AMAT} = 2 + 1.00 \times 2 = 4.0 \text{ cycles}$$

With prefetcher:

$$\text{AMAT} = 2 + 0.20 \times 2 = 2.4 \text{ cycles}$$

C. Prefetch Accuracy

17 prefetch requests were issued for 16 hits. One extra prefetch may be generated speculatively beyond the final CPU access (for address 0×143), which is never demanded by the CPU. This represents normal prefetcher behavior—the overprediction rate is negligible and has no correctness impact.

D. Waveform Observations

From the XSim waveforms:

- **TB1 (Fig. 1a):** The `busy` signal pulses for exactly 4 cycles on each miss ($S1 \rightarrow S2 \rightarrow S3 \rightarrow S0$) and 2 cycles on each hit ($S1 \rightarrow S0$). The `hit` signal is asserted combinationally in $S1$ and is stable before the FSM transitions.
- **TB2 (Fig. 1b):** Multiple cache sets are exercised; the address bus shows the expected patterns including LRU eviction verification (re-access of surviving ways returns hits).
- **TB3 (Fig. 1c):** `hit` remains deasserted throughout all 20 accesses. `prefetch_req` stays at 0 (prefetcher bypassed). AMAT and rate signals all register as expected at simulation end.
- **TB4 (Fig. 1d):** `prefetch_req` pulses are visible between CPU accesses (during the 10-idle-cycle windows). The transition from misses to hits (access 5 onward) is reflected in the `hit` signal. `total_hits` and `total_misses` stabilise at 16 and 4 respectively.

V. KEY DESIGN CHALLENGES

Several non-trivial RTL issues arose during implementation:

1. LRU Counter Wrap-around. Using non-blocking assignments naively caused the age of the *victim* to be read before it was written. The fix: age only ways with `lru < lru[hit_way]` (only more-recently-used peers), avoiding cascaded increments.

2. Edge Detection in Prefetcher. `cs` and `rd` remain high for the full 4-cycle miss duration. Without a `prev_cs` edge detector, the prefetcher sampled the same address on every clock cycle and computed a false stride of zero. The `!prev_cs` guard limits processing to the first cycle of each CPU request.

3. Premature Prefetch Deassert. Without the two-phase handshake, `prefetch_req` could be cleared before the cache FSM had latched the prefetch address, causing prefetches to be silently dropped. The `wait_for_busy` flag synchronises the clear to the cache’s acceptance of the request.

4. Testbench cs/rd Deassert Timing. Deasserting `cs/rd` must occur *immediately* after `wait(busy==0)` with no intervening clock edge. Any delay allowed the FSM to observe `cs && rd` again in $S0$ and launch a duplicate transaction.

5. LRU Eviction Test Ordering. In the eviction testbench, hit verification for surviving ways must be performed *before* re-loading the evicted block. Doing so afterward risked cascade eviction displacing additional valid ways and producing misleading results.

VI. CONCLUSION

This paper presented a complete RTL implementation of a prefetch-aware L1 cache management system in Verilog, simulated and verified in Vivado XSim. The 4-way set-associative cache with LRU replacement correctly handles cold misses, hits, and LRU evictions as confirmed by four targeted testbenches. The integrated stride prefetcher detects regular access patterns within 3 accesses and begins prefetching on the 4th, converting subsequent misses into hits.

The key quantitative result is a **40% reduction in AMAT** ($4.0 \rightarrow 2.4$ cycles) for a representative sequential workload, achieved with no change to the CPU interface and at the cost of a 4-entry prefetcher table and a combinational arbiter.

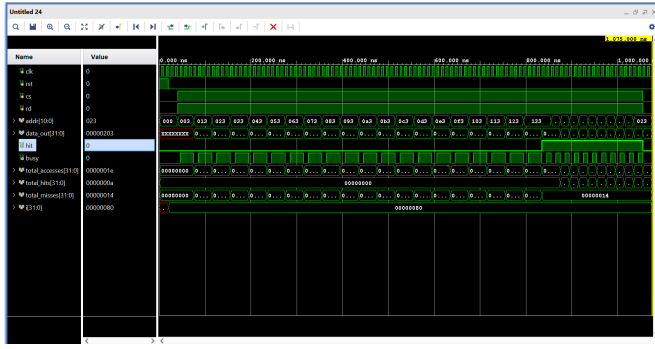
Future work includes extending the prefetcher to handle multiple simultaneous strides, implementing a write-back policy in the cache, and synthesising the design onto an FPGA to measure real timing margins.

ACKNOWLEDGMENT

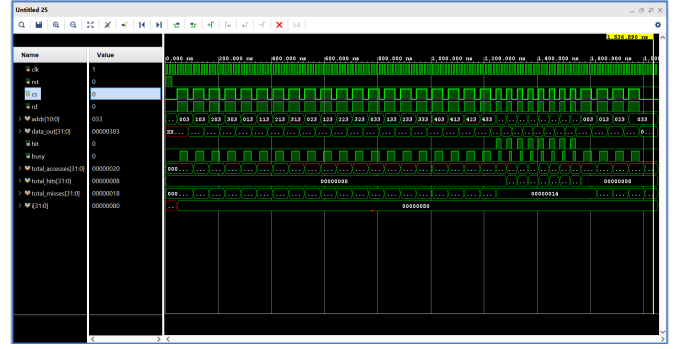
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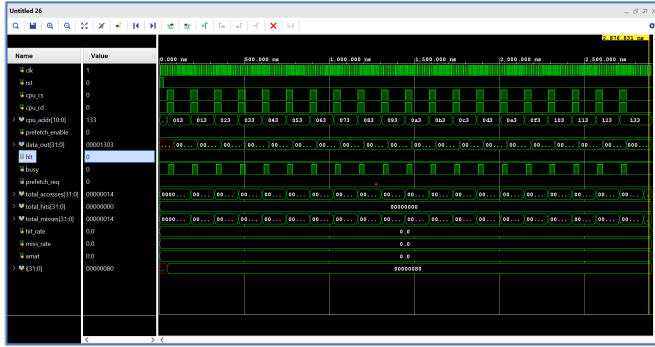
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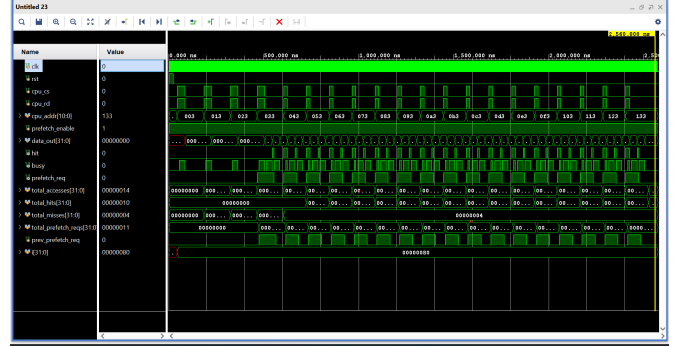
(a) TB1: Cold miss and repeated hit test.
total_accesses=30, hits=10, misses=20.



(b) TB2: LRU eviction test.
total_accesses=32, hits=8, misses=24.



(c) TB3: Baseline (no prefetcher).
20 accesses, 0 hits, 20 misses, AMAT = 4.0 cycles.



(d) TB4: With stride prefetcher.
20 accesses, 16 hits, 4 misses, AMAT = 2.4 cycles.

Fig. 1: XSim simulation waveforms for all four testbenches.

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